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9702/32

May/June 2007

2 hours

Additional Materials: As listed in the Confidential Instructions.

READ THESE INSTRUCTIONS FIRST

DO NOT WRITE ON ANY BARCODES.

You are reminded of the need for good English and clear presentation in your answers.

All questions in this paper carry equal marks.

For Examiner's Use	
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2	
Total	

This document consists of **8** printed pages and **4** blank pages.

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You may not need to use all of the materials provided.

- 1** In this experiment you will measure the current I through a resistor R_3 as its resistance is changed.

(a) (i) Use the voltmeter to measure the e.m.f. E of the power supply.

$E = \dots\dots\dots$ V

- (ii)** Connect the circuit shown in Fig. 1.1. R_1 and R_2 are labelled, and R_3 may be chosen from any one of the remaining resistors. Each resistor carries a label indicating its resistance.

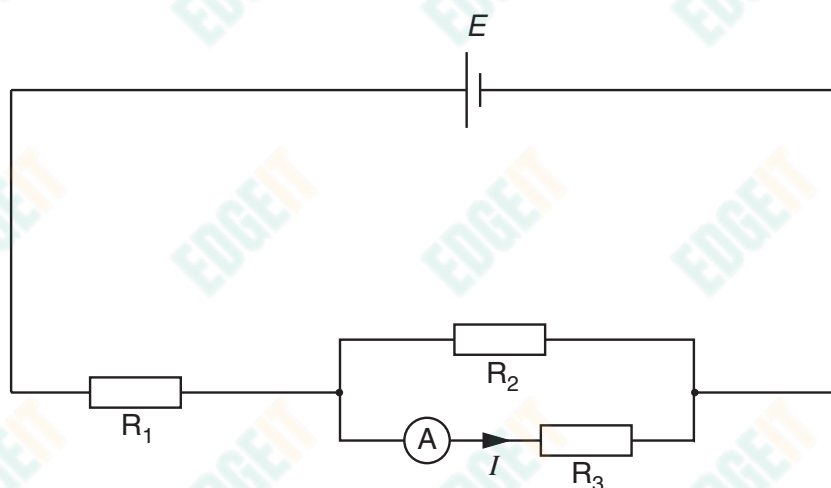


Fig. 1.1

- (iii)** Record the values of the current I and the resistance R_3 of resistor R_3 .

$I = \dots\dots\dots$ A

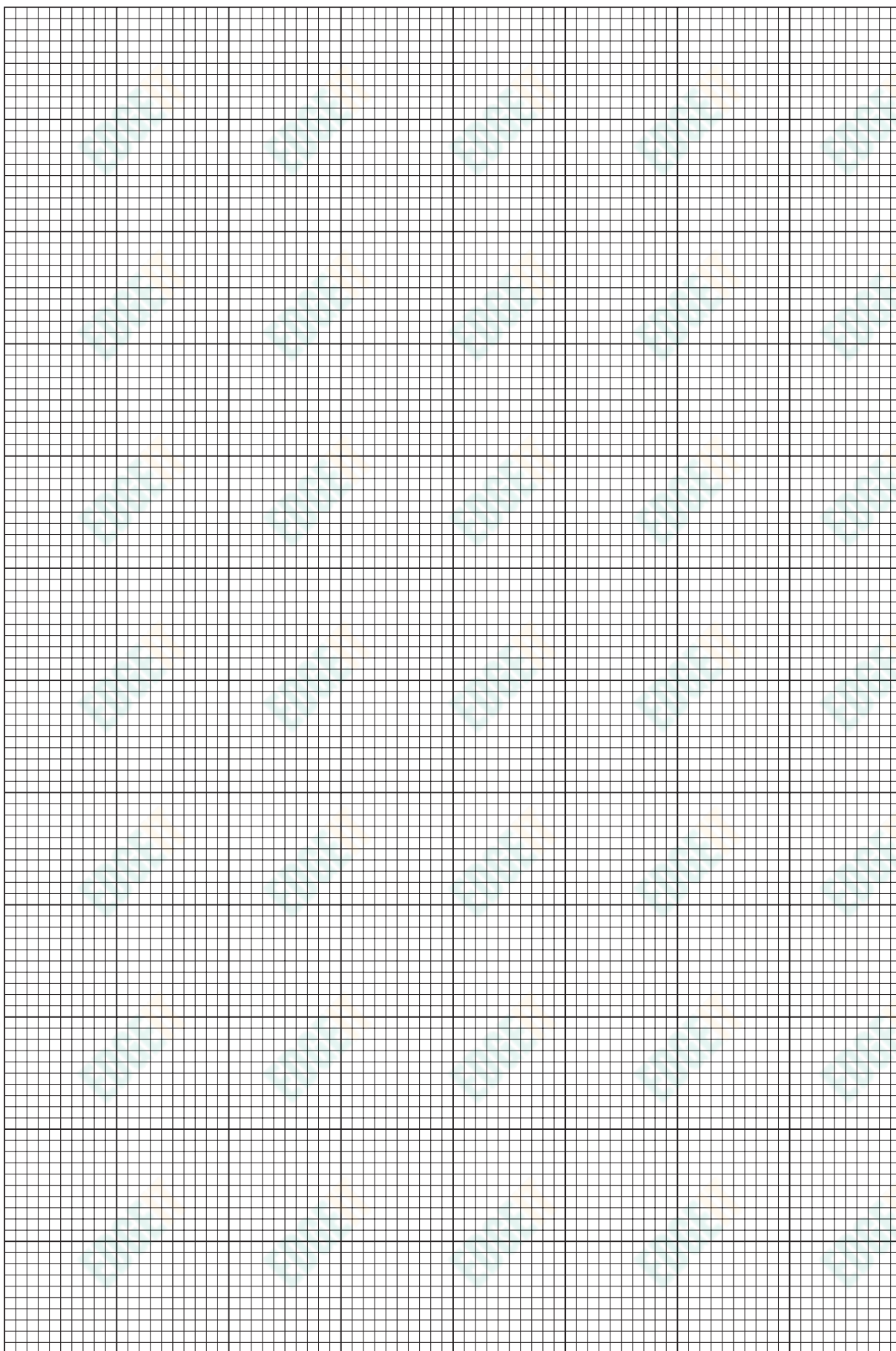
$R_3 = \dots\dots\dots$ Ω

- (b) Change the resistor R_3 and repeat (a)(iii) until you have six sets of readings for I and R_3 . Include values of $\frac{1}{I}$ in your table of results.

- (c) (i) Plot a graph of $\frac{1}{I}$ (y -axis) against R_3 (x -axis).
(ii) Draw the line of best fit.
(iii) Determine the gradient and the y -intercept of the graph.

gradient =

y -intercept =



(d) The relationship between I and R_3 is

$$\frac{1}{I} = \left(\frac{R_1 + R_2}{ER_2} \right) R_3 + \frac{R_1}{E}$$

where R_1 is the resistance of the resistor R_1 , R_2 is the resistance of the resistor R_2 , and E is the e.m.f. of the power supply.

Using your answers from (a)(i) and (c)(iii), determine values of R_1 and R_2 .

$R_1 = \dots\dots\dots$

$R_2 = \dots\dots\dots$

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You may not need to use all of the materials provided.

- 2** In this experiment you will investigate how the rebound height h of a table-tennis ball is related to the height d from which it is dropped.

If air resistance is ignored, then theory predicts that

$$e = \sqrt{\frac{h}{d}}$$

where e is a constant. Air resistance may be ignored if $d < 40$ cm.

- (a)** **(i)** Mount a half-metre rule vertically using a stand, boss and clamp.
(ii) Place the table-tennis ball next to the rule and record d . Release the ball and make measurements to determine the rebound height h , as shown in Fig. 2.1.

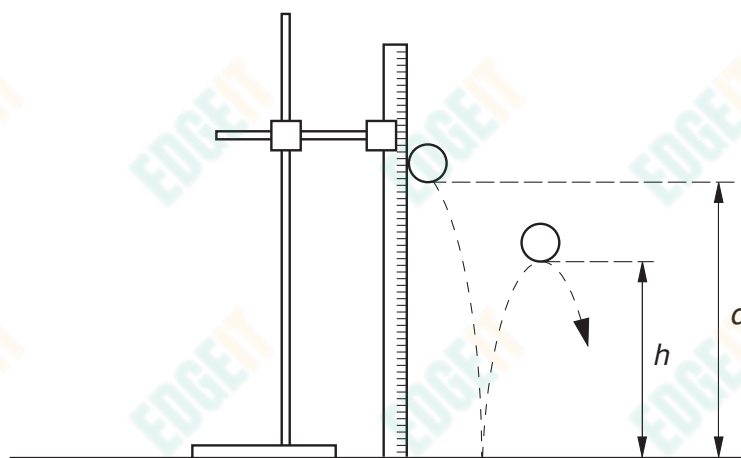


Fig. 2.1

$d =$

$h =$

- (iii)** Explain how you used the apparatus to ensure that the rebound height h was measured as accurately as possible.

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- (iv) Estimate the percentage uncertainty in h .

percentage uncertainty in $h = \dots\dots\dots$

- (v) Calculate a value for e .

$e = \dots\dots\dots$

- (b) Repeat (a)(ii) and (b)(v) using a different value of d .

$d = \dots\dots\dots$

$h = \dots\dots\dots$

$e = \dots\dots\dots$

- (c) Do the results of your experiment indicate that e is a constant within the limits of experimental accuracy? Explain your reasoning clearly.

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(d) (i) State four sources of error or limitations of the procedure in this experiment.

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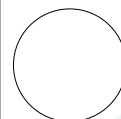
(ii) Suggest four improvements that could be made to this experiment. You may suggest the use of other apparatus or different procedures.

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